

JAYPEE UNIVERSITY OF ENGINEERING & TECHNOLOGY, GUNA
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Course: Computer Organization & Architecture Lab

Course Code: CS208

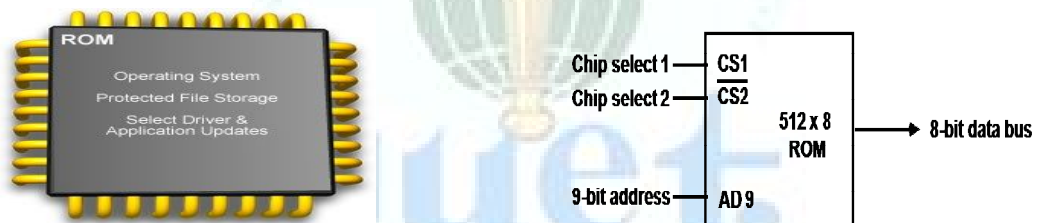
B. Tech. (CSE IV Sem.)

Experiment # 10

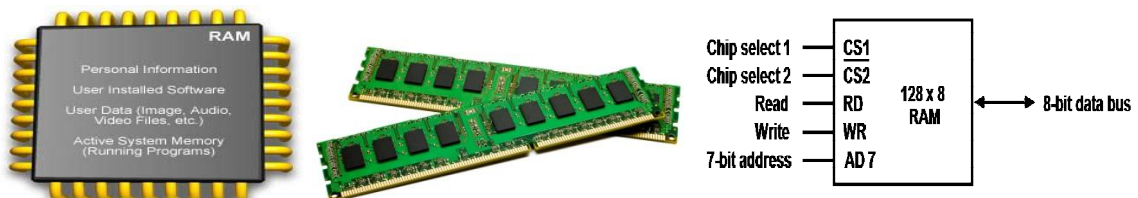
Aim: Design of ROM and RAM.

The memory unit is an essential component in any digital computer and it is needed for storing programs and data. The memory unit that communicates directly with the CPU is called the **main (primary) memory**. Devices that provide backup storage are called **auxiliary (secondary) memory**. The most common auxiliary memory devices used in computer systems are magnetic disks and tapes. They are used for storing system programs, large data files, and other backup information. Programs and data currently needed by the processor reside in main memory. There are two types of main memories:

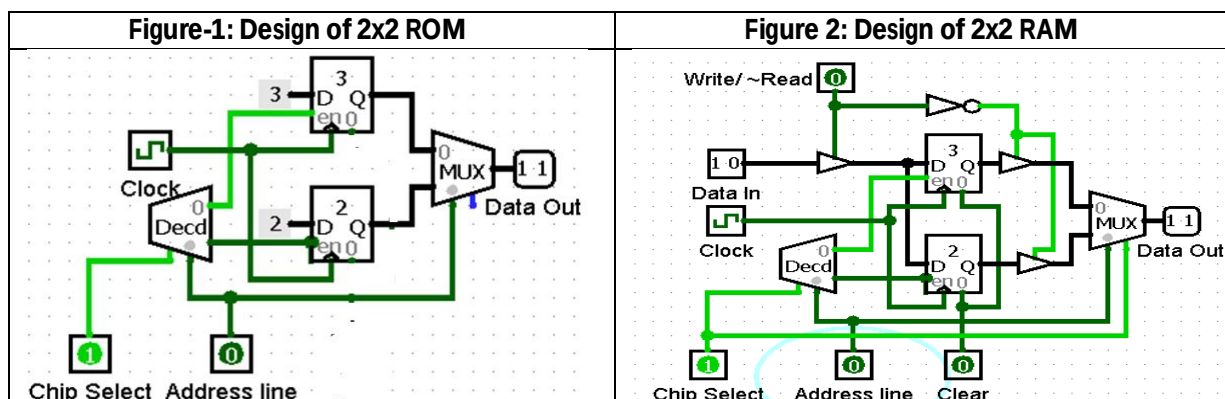
Read Only Memory (ROM): This is the **read only** memory which is used for storing programs that are permanently resident in the computer and for tables of constants that do not change in value. It requires **unidirectional data bus** to transfer data from memory to CPU. This memory also stores the initial programs (BIOS program, operating system) whose function is to boot the computer when power is turned on. This memory comes in category of **non-volatile** memory because the contents of ROM remain unchanged after power is turned off and on again.



Random Access Memory (RAM): This is the **read and write** memory which is used for storing user running programs, user data, and user installed software etc. A RAM chip is used for communication with the CPU using **bidirectional data bus** that allows the transfer of data either from memory to CPU during a read operation or from CPU to memory during a write operation. This memory comes in category of **volatile** memory because the contents of RAM get destroyed after power is turned off.



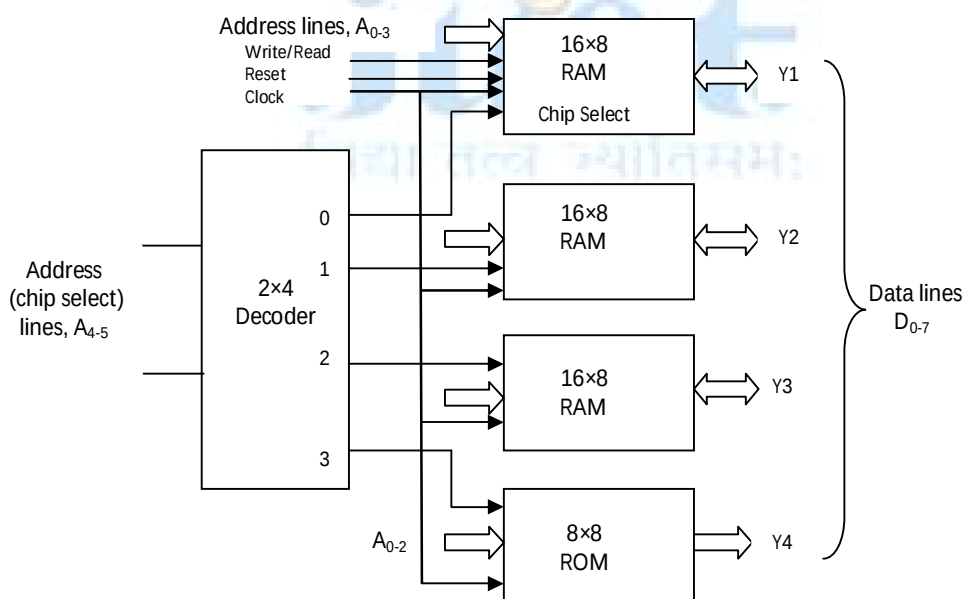
Example: Design of ROM and RAM using **in-built block** of register.



Exercise#1: Design 8x8 ROM (convert Fig. 1) and 16x8 RAM (convert to Fig. 2) using **in-built block** of register. Perform Write/Read operation in the memories using following in instructions:

Contents of ROM			Contents of RAM		
Address	Symbol	Hex Code	Address	Symbol	Hex Code
1	CIR	78	0	LDA	20
3	CIL	7F	3	ADD	13
4	HLT	70	A	CLA	78
6	BUN	5B	D	STA	3A
7	AND	04	E	INC	7D

Exercise#2: Design 56x8 memory system as shown in block diagram using **sub-circuits** of RAM and ROM chips designed in Exercise#1. Find the address maps of each memory chip and perform Write/Read operation in the memories using instructions given in the table:



Contents of Memory			
Chip Select bits	Address bits	Symbol	Hex Code
00	0101	LDA	20
00	1101	ADD	13
01	1001	CLA	78
01	0101	STA	3A
10	1110	INC	7D
10	0000	BUN	5B
11	X111	CIL	7F
11	X100	HLT	70

X = don't care condition

Exercise#3: Repeat **Exercise#2** using in-built blocks of RAM and ROM memories.

[Note: Go to **RAM** → **Data Interface** → **Separate load and Store ports** to enable data input terminal (D) and Write (str) / Read (ld) control lines.]

Exercise#4: Design 384×8 memory system using **in-built blocks** of RAM of size 256×8 and ROM of size 128×8. Use **in-built blocks** of **counter** for auto write of data in RAM and auto address updates of both the memories. Perform following operations in the designed system:

- Enter Hex ASCII codes of your name in ROM.
- Display the printing of your name using **in-built block** of **TTY** (teletypewriter).
- Replace counter block with **in-built block** of **Random Generator** to display the printing of ASCII codes in random order.

Design Steps:

- Use inbuilt block of RAM with data and address of 8 bits.
- Use inbuilt block of ROM with 8 data bits and 7 address bits.
- Use one 1: 2 decoder for chip selection.
- Use two 8-bit counters, first counter for auto address update of both the memory chips and second counter for auto data write in RAM.
- Use TTY for auto typing the contents of memory chips.
- Apply common clock signal in all the blocks and make other connections as per requirements.

ASCII Table

Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	\$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]